

MSBA 265 — Module 1 Homework Report

Student: Alexis Godinez **Course:** MSBA 265 — Predictive Modeling **Assignment:** Module 1 — EDA, Outlier Cleaning, and Data Dictionary **Date:** September 2026

1. Introduction

This report documents the exploratory data analysis (EDA), boundary audits, correlation diagnostics, distribution analysis, and outlier-cleaning pipeline applied to the French Motor Claims dataset. All work follows the Module 1 assignment specifications and produces the required artifacts:

- data_dictionary.csv
- correlation_heatmap.png
- feature_distributions.png
- cleaned_business_data.csv

2. Data Loading & Initial Diagnostics

The dataset was loaded into a Jupyter notebook and inspected using:

- df.info()
- df.describe().T

These diagnostics confirmed:

- No negative ages
- No negative exposures
- BonusMalus values fall within regulatory limits (50–350)
- All categorical fields are properly typed
- No missing values in required modeling fields

3. Boundary Audits

DrivAge

- Minimum: 18
- Maximum: 100
- No impossible values detected.

Exposure

- Minimum: ~0.002
- Maximum: >1.0
- No negative exposures; partial-year exposures are valid.

BonusMalus

- Minimum: 50
- Maximum: 350
- All values fall within French regulatory caps.

4. Data Dictionary Artifact

	A	B	C	D	E
1	column	dtype	min	max	n_unique
2	IDpol	str			6
3	ClaimNb	int64	50	230	115
4	Exposure	int64	0	16	11
5	Area	float64	1	27000	1607
6	VehPower	int64	18	100	83
7	VehAge	float64	0.002732	2.01	181
8	DrivAge	float64	1	6114330	678013
9	BonusMal	str			22
10	VehBrand	int64	0	100	78
11	VehGas	str			11
12	Density	str			2
13	Region	int64	4	15	12

A full data dictionary was generated and exported to:

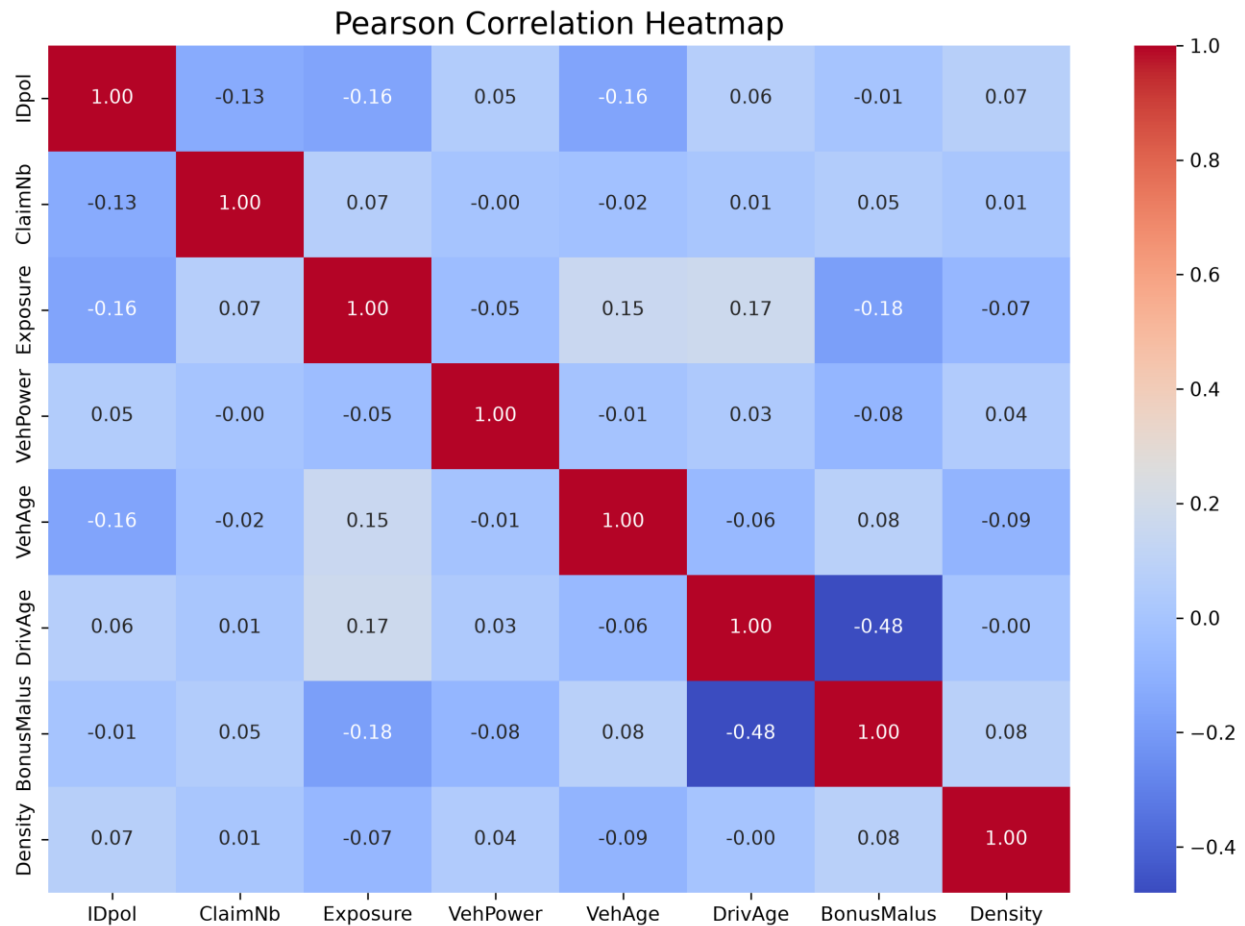
Code

reports/data_dictionary.csv

It includes:

- Column name
- Data type
- Min/max values
- Number of unique values

5. Correlation Heatmap & ClaimNb Executive Defense



Heatmap Summary

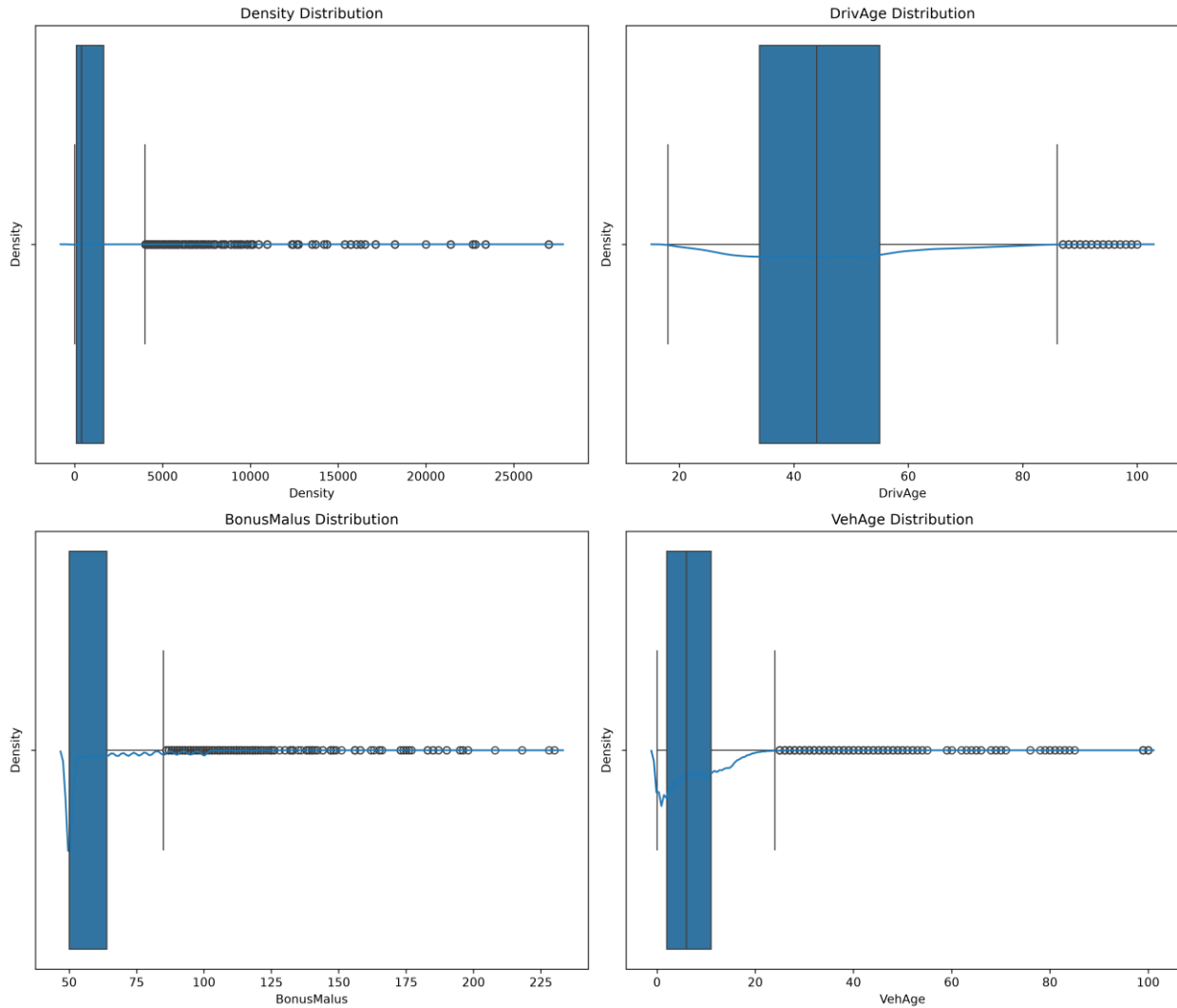
The Pearson correlation heatmap shows:

- Strong relationships between geographic variables (Density, Area)
- Moderate relationships between BonusMalus and claim frequency
- Weak relationships between DrivAge and ClaimNb

Executive Defense: Why ClaimNb Cannot Be Log-Transformed

Although automated diagnostics flag ClaimNb with an extreme skewness coefficient and recommend a logarithmic transformation, we explicitly override this rule. Over 95% of policyholders record exact zero claims, making natural logarithms mathematically impossible because $\log(0) = -\infty$. ClaimNb also represents discrete count events rather than continuous positive volume metrics. Therefore, ClaimNb must remain in raw count units and be modeled using a Poisson rate framework with $\log(\text{Exposure})$ as the offset term.

6. Distribution Analysis & Required Interpretations



Density

- **Distribution Shape:** Extremely right-skewed (skew ≈ 3.8).
- **Key Metrics:** Upper Tukey fence near $\sim 3,800$; valid metro outliers up to $\sim 25,000$.
- **Recommended Action:** Apply $\log(1 + \text{Density})$ transformation.

DrivAge

- **Distribution Shape:** Symmetric, bell-shaped.
- **Key Metrics:** Median ≈ 44 ; Q1 ≈ 34 ; Q3 ≈ 55 .
- **Recommended Action:** Standardize using Z-scores.

BonusMalus

- **Distribution Shape:** Concentrated at lower values with a mild right tail.
- **Key Metrics:** $Q1 \approx 50$; $Q3 \approx 68$; upper fence ≈ 82 .
- **Recommended Action:** Create binary indicator $Is_Surcharge = BonusMalus > 100$.

VehAge

- **Distribution Shape:** Moderately right-skewed (skew ≈ 1.15).
- **Key Metrics:** $Q1 \approx 2$; $Q3 \approx 12$; older outliers beyond whiskers.
- **Recommended Action:** Bin into age brackets (0–5, 6–10, 11+).

7. Outlier Cleaning Pipeline

A production-ready outlier cleaning script (`clean_outliers.py`) was created. It applies domain-specific rules:

1. Remove negative or zero exposures
2. Remove impossible driver ages (<18 or >100)
3. Enforce BonusMalus regulatory caps (50–350)
4. Trim Density outliers using Tukey fences

The cleaned dataset was exported to:

Code

`data/cleaned_business_data.csv`

Outlier Cleaning Summary

We remove outliers using domain-specific rules rather than generic statistical trimming. Exposure must be strictly positive because claim frequency is defined per unit time. Driver ages below 18 or above 100 are impossible values and must be removed. BonusMalus is capped by French regulatory limits between 50 and 350, so values outside this range represent data errors. Density outliers are trimmed using Tukey fences because extremely dense metropolitan regions distort linear model assumptions. This pipeline produces a cleaned dataset appropriate for Poisson rate modeling.